

Digital Forensics Laboratory

EXPERIMENT NO: 05

DATE: August 6, 2026

TITLE: Digital Forensics Investigation Using Autopsy: Case Creation and Evidence Import

1. AIM

To use the Autopsy digital forensics platform to create a new forensic case, import digital evidence (disk image files), configure ingest modules, and perform basic forensic analysis and reporting.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Forensics Tool:** Autopsy Digital Forensics Platform
- **Evidence Files:** Disk image files (4Dell Latitude CPi.E01, 4Dell Latitude CPi.E02) downloaded from the provided forensic repository.

3. THEORY

Autopsy is a premier open-source digital forensics platform and graphical interface to The Sleuth Kit® and other digital forensics tools. It is widely used by law enforcement, military, and corporate examiners to analyze and extract critical data from digital devices.

In a standard forensic investigation, raw data or disk images (such as .E01, .dd, or .raw formats) are imported into a securely managed "Case". Autopsy utilizes various **Ingest Modules** (like File Type Identification, Keyword Searching, Timeline Analysis, and Hash Lookups) to process this evidence in the background, automatically categorizing artifacts such as web history, file system metadata, and communication records for the examiner to review.

4. PROCEDURE

Step 1: Installation & Preparation

1. Download and install the Autopsy application based on the host operating system.
2. Download the required evidence files (4Dell Latitude CPi.E01) to a secure local directory.
3. Launch the Autopsy application to view the welcome screen.

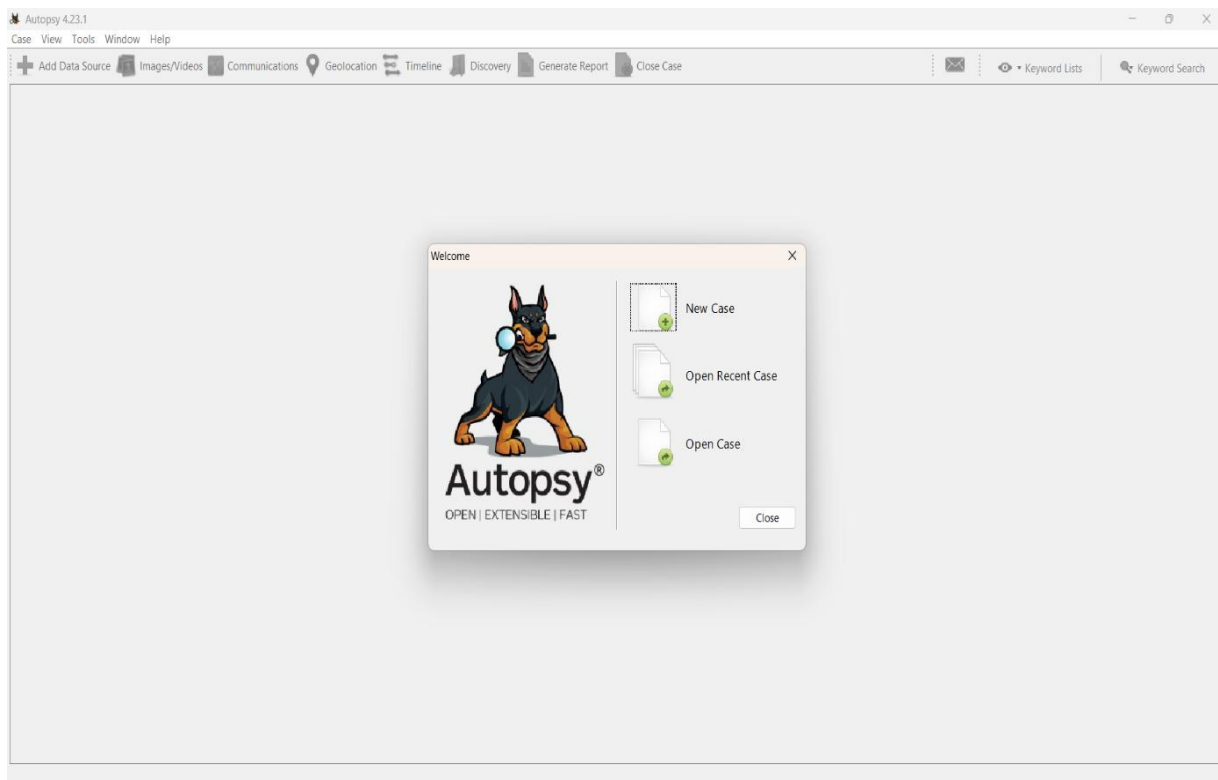


Figure 1: Autopsy startup menu and tool interface

Step 2: Starting a New Case

1. Click on **New Case** from the startup menu.
2. Enter the case name (e.g., Inba_Investgation) and specify the base directory where the case data will be stored.
3. Click **Next** to proceed to the optional information.

4. Fill in the required case details, including the case number and examiner's name (Inbathamizhan S), then click **Finish**.

New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number: Case - 2

Examiner

Name: M.reddappa Reddy

Phone: 6385631533

Email: 99240040292@klu.ac.in

Notes:

Organization

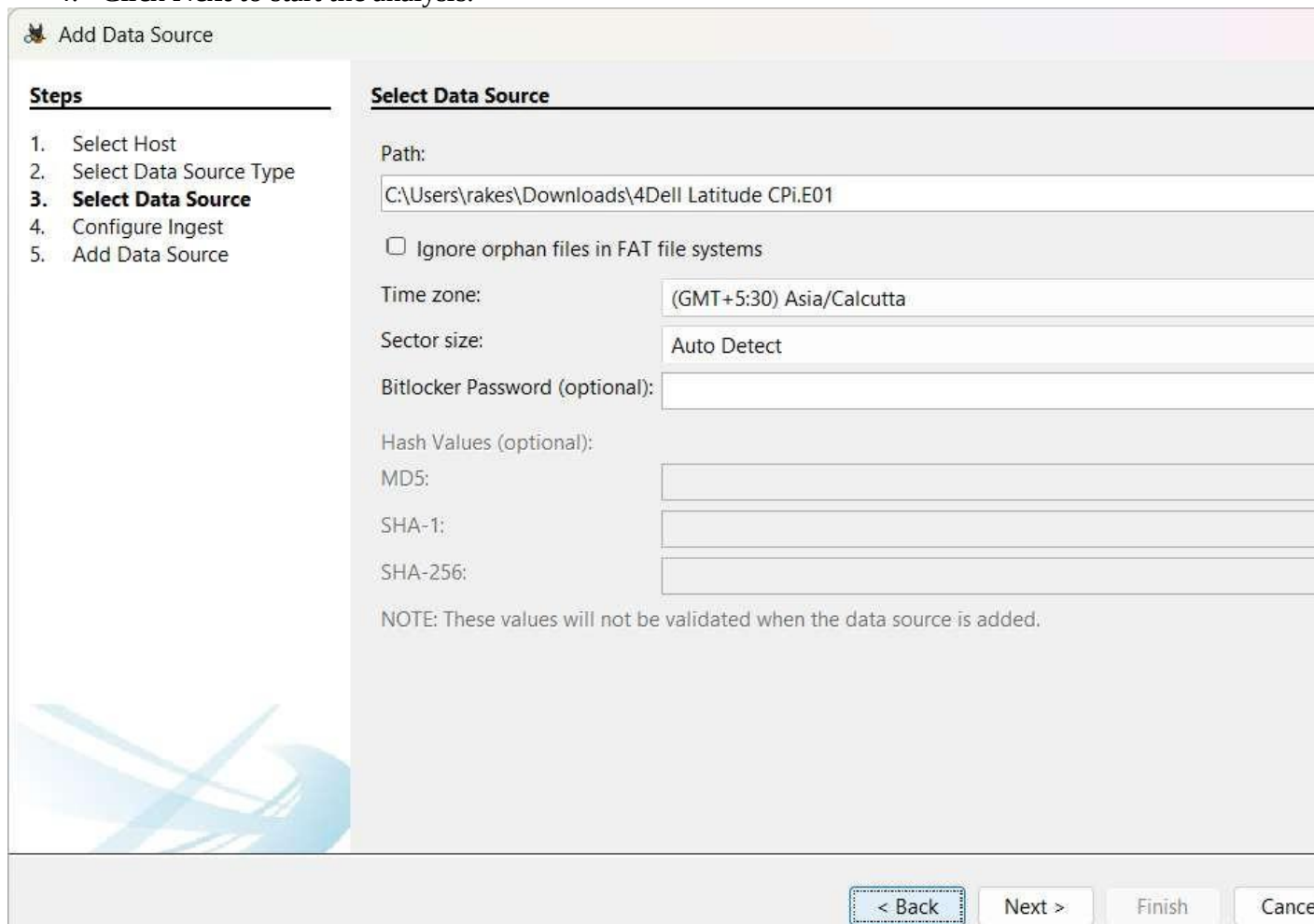
Organization analysis is being done for: Not Specified

< Back Next > **Finish** Cancel Help

Figure 3: Entering Optional Information including Examiner details

Step 3: Adding a Data Source

1. When prompted to add a data source, choose the type of data source (e.g., Disk Image or VM file).
2. Browse to the local directory and select the downloaded evidence file (4Dell Latitude CPi.E01).
3. **Configure Ingest Modules:** Select the specific analysis modules required for the investigation.
4. Click **Next** to start the analysis.



The screenshot shows a software window titled "Add Data Source". On the left, a "Steps" sidebar lists five steps: 1. Select Host, 2. Select Data Source Type, 3. **Select Data Source** (highlighted), 4. Configure Ingest, and 5. Add Data Source. The main area is titled "Select Data Source" and contains the following fields and options:

- Path:** A text box containing "C:\Users\rakes\Downloads\4Dell Latitude CPi.E01".
- ☐ Ignore orphan files in FAT file systems
- Time zone:** A dropdown menu showing "(GMT+5:30) Asia/Calcutta".
- Sector size:** A dropdown menu showing "Auto Detect".
- Bitlocker Password (optional):** An empty text box.
- Hash Values (optional):** A section with three empty text boxes labeled "MD5:", "SHA-1:", and "SHA-256:".

Below these fields, a note states: "NOTE: These values will not be validated when the data source is added." At the bottom right, there are four buttons: "< Back" (highlighted with a dashed border), "Next >", "Finish", and "Cancel".

laptop theft - Autopsy 4.23.1

CaseViewToolsWindowHelp

Add Data Source

Images/Videos

Communications

Geolocation

Timeline

Discovery

Generate Report

Close Case

Keyword Lists

Data Sources

4Dell Latitude CPi.E01_1 Host

4Dell Latitude CPi.E01

File Views

File Types

Deleted Files

MB File Size

Data Artifacts

Metadata (19)

Operating System Information (1)

Recent Documents (8)

Run Programs (59)

Web Bookmarks (6)

Web Cookies (24)

Web History (18)

Analysis Results

Encryption Suspected (1)

Extension Mismatch Detected (8)

Interesting Items (4)

Web Categories (3)

OS Accounts

Tags

Score

Reports

Listing

4Dell Latitude CPi.E01_1 Host

TableThumbnailSummary

Name	Type	Size (Bytes)	Sector Size (Bytes)	Timezone	Device ID
4Dell Latitude CPi.E01	Image	4871301120	512	Asia/Calcutta	8e29c91c-afdc-452b-9a43-b1250f63d835

HexTextApplicationFile-MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

Analyzing files from 4Dell Latitude CPi.E01100%

Step 6: Reporting and Case Closure

1. Click on **Generate Report** from the main toolbar.
2. Choose the desired report format (e.g., Excel Report, HTML Report) under the Select and Configure Report Modules window.
3. Click **Next** and select the specific result types to include (e.g., Accounts, E-Mail Messages, etc.).
4. Review the generated report to ensure all relevant information and exported artifacts are accurately documented.
5. Once the investigation is complete, close the case within Autopsy and archive all data according to organizational policies.

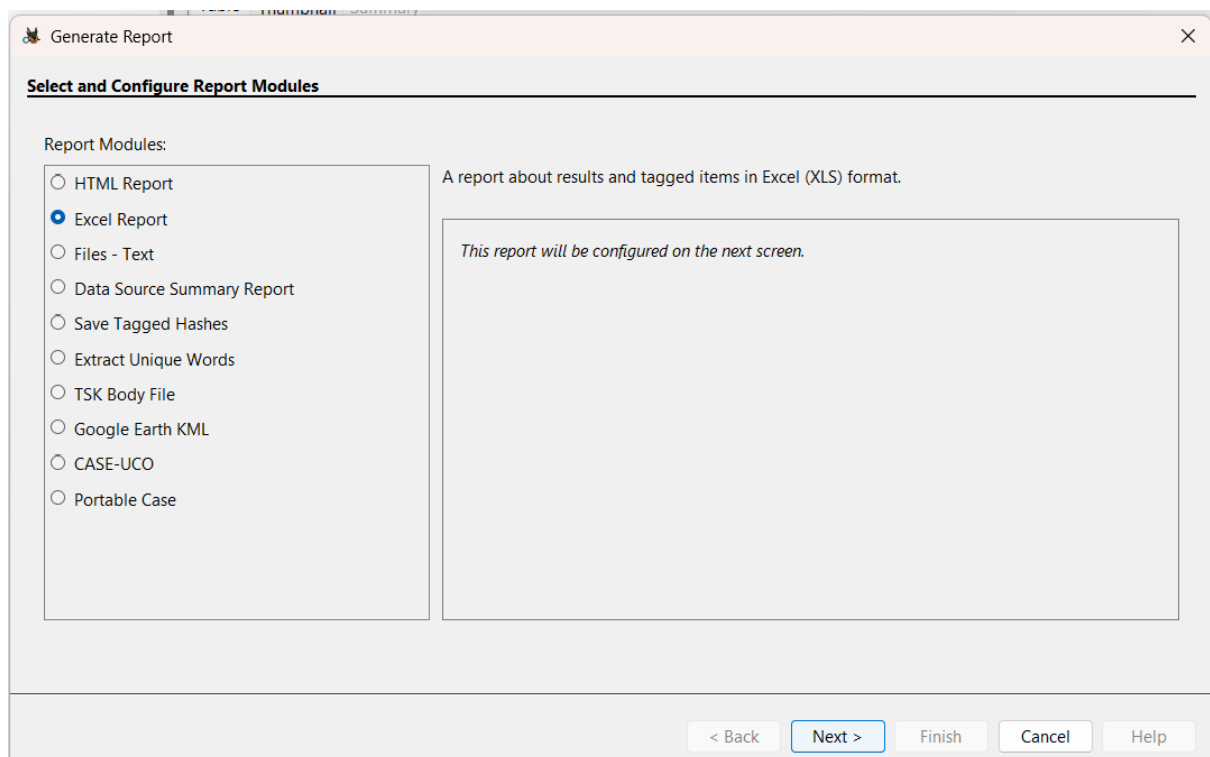


Figure 7: Selecting the report module format

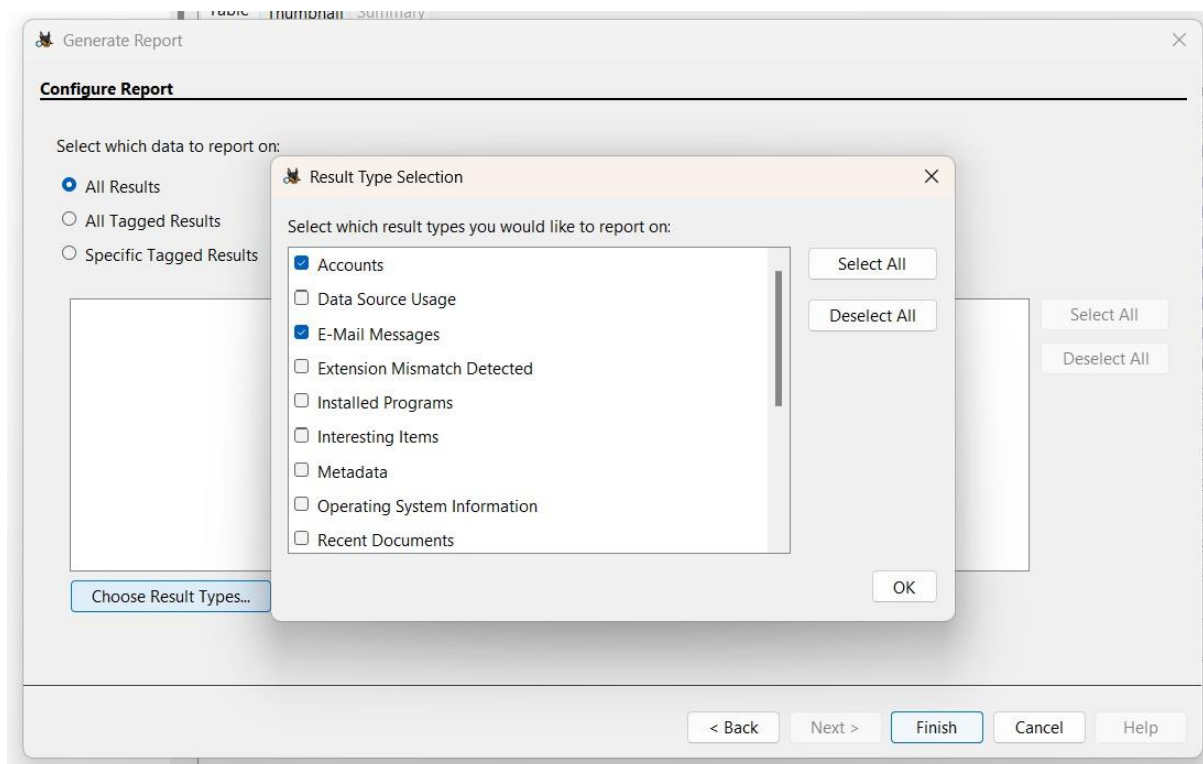


Figure 8: Configuring the report by selecting specific tagged data

5. RESULT

The digital forensics experiment was successfully completed. A new case was established in Autopsy, and the provided .E01 disk image evidence was successfully imported, analyzed using ingest modules, and documented via the reporting feature.

Autopsy Forensic Report

Warning, this report was run before ingest services completed!

HTML Report Generated on 2026/07/28 10:29:15

Case:	laptop theft
Case Number:	case- 001
Number of data sources in case:	1
Examiner:	Rakesh

Image Information:

4Dell Latitude CPi.E01

Timezone:	Asia/Calcutta
Path:	C:\Users\rakes\Downloads\4Dell Latitude CPi.E01

Software Information:

Autopsy Version:	4.23.1
Android Analyzer Module:	4.23.1

Complete

HTML Report : D:\laptop theft\Reports\laptop theft HTML Report 07-28-2026-10-29-15\report.html

Complete

Cancel

Close